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github/Gavino3o

Corp Governance and Agency Prob

Investment vs Financing Decisions

- **Investment Decision**: How to allocate capital to long-term assets (tangible/intangible assets)
- Also known as **Capital Budgeting Decision** or **Capital Expenditure (CAPEX) Decision**
- Examples of investment decisions: launching a new product, expanding into a new market, acquiring another company, investing in research and development
- **Tangible assets**: physical assets used in operations.
- **Intangible assets**: non-physical assets that provide long-term value.
- Examples of tangible assets : machinery, buildings, land
- Examples of intangible assets: patents, trademarks, copyrights, goodwill
- **Real assets**: physical assets used in operations (e.g. machinery, buildings, land)
- **Financial assets**: financial instruments that represent a claim on future cash flows (e.g. stocks, bonds, derivatives)

- **Financing Decision**: How to raise capital to fund long-term investments
- Also known as **Capital Structure Decision**
- Mixture of debt and equity financing
- Debt financing: borrowing money from external sources (e.g. banks, bondholders) with the obligation to repay with interest
- Equity financing: issuing shares of stock to investors in exchange for ownership stake in the company

Corporation

- A business organised as a separate legal entity owned by shareholders
- Shareholders elect a board of directors, who appoints managers to run the company and monitor their performance. Seperation of ownership and control.
- **Financial Managers:** CFO, Treasurer, Controller
- (**Permanence**): Corporation continues to exist even if ownership changes
- (**Limited Liability**): The owners of a corporation are not personally liable for its obligations. Shareholders are only liable for the amount they invested in the company.
- **Types of Corporations**:
  - **Public Corporation**: owned by a large number of shareholders and traded on public stock exchanges
  - **Private Corporation**: owned by a small group of shareholders and not traded on public stock exchanges
- **Types of Business Organisations**:
  - **Sole Proprietorship**: owned and operated by a single individual
  - **Partnership**: owned and operated by two or more individuals who share profits and losses
  - **Corporation**: separate legal entity owned by shareholders
  - **Limited Liability Company (LLC)**: hybrid structure that combines the benefits of a corporation and a partnership. No double taxation, limited liability, flexible management structure.
  - **Professional Corporation (PC)**: for licensed professionals (e.g. doctors, lawyers, accountants). Limited liability for business debts, but not for professional malpractice (can be sued).

	Sole Proprietorship	**Partnership**	**Corporation**
Who owns the business?	The manager	Partners	Stockholders
Are managers and owners separate?	No	No	Usually
What is the owner's liability?	Unlimited	Unlimited	Limited
Are the owner and business taxed separately?	No	No	Yes

Agency Problem

- **Goals of a Corporation:**
  - Shareholders desire wealth maximization. Maximizing the current market value of shareholder's investment in the company.
  - Why not profit maximization? Profit is a flow variable, while wealth is a stock variable. Profit does not account for the time value of money, risk, and cash flows. Earning manipulation to increase short-term profit may harm long-term wealth.
  - **Opportunity cost of capital**: minimum acceptable rate of return on capital investments by shareholders (set by investment alternatives in the financial markets)
- **Agency Problem**: conflict of interest between shareholders (principals) and managers (agents). Managers are agents of shareholders and are tempted to act in their own interests rather than maximizing shareholder value.
- Managers have constituencies called **stakeholders** (e.g. employees, customers, suppliers, creditors, government) whose interests may not align with shareholders.
- **Agency Costs:** Value lost from agency problems or from efforts to mitigate agency problems.
- **Solutions to Agency Problem:**
  - **Executive Compensation**: align manager's interests with shareholders. E.g. stock options, bonuses tied to performance metrics (e.g. stock price, earnings per share)
  - **Corporate Governance**: Systems of laws, regulations, policies, institutions, and corporate practices that protect shareholders and other investors.
  - **Elements of Good Corporate Governance**:
    - Legal requirements and regulations
    - Board of Directors: independent directors, committees (e.g. audit, compensation, nomination)
    - Activist shareholders: large shareholders who actively monitor and influence management
    - Takeovers: threat of takeover can discipline management (e.g. If you share price is undervalued, another company may acquire you and replace management)
    - Transparency and disclosure: accurate and timely financial reporting, disclosure of material information

Ethics of Maximising Shareholder Wealth

- **Short selling**: selling borrowed shares in the hope of buying them back at a lower price to make a profit. (Can lead to market manipulation)
- **Corporate raiders**: investors who acquire a large stake in a company to influence management and increase shareholder value.
- **Tax avoidance**: legal strategies to minimize tax liability (e.g. offshore accounts, tax shelters)

Platform Business Models

- **Linear Value Chain:** traditional business model where value is created through a series of sequential activities (e.g. Design $\rightarrow$ Manufacture $\rightarrow$ Sell $\rightarrow$ Deliver)
- **Platform Business Model:** business model that creates value by facilitating exchanges between two or more interdependent groups (e.g. buyers and sellers, users and developers)
  (Producers $\rightarrow$ Platform $\leftarrow$ Consumers)
- Examples: Uber, Airbnb, Amazon, Facebook, Google

Types of Platforms:

- **Product Platforms**: provide a foundation for creating and distributing products or services with improvements over time (e.g. iPhone, Tesla)
- **Transaction Platforms**: facilitate exchanges between buyers and sellers (e.g. eBay, Uber, Airbnb)
- **Innovation Platforms**: provide a foundation for third-party developers to create complementary products or services (e.g. iOS, Android, Windows)
- **Orthogonal Platforms**: Products or services that are based on the sale of different services of two groups of customers who do not come into contact with each other (e.g. newspaper adverstising [client as a target], data brokers [client as a source])
- **Hybrid/Integrated Platforms**: combine elements of transaction and innovation platforms (e.g. Amazon, Google)

Advantages of Platform Business Models:

- No gatekeepers: anyone can participate as a producer or consumer. Consumers have more choices and producers have access to a larger market.
- New sources of value and supply: Asset owning is not necessary. Expansion is less risky. Sharing economy (Optimising the use of idle assets)
- Community feedback loops: users can provide feedback and ratings, which can improve the quality of products and services.
- Focus has shifted from broadcast to segmentation, and then to virality and social influence.
- From push to pull: consumers have more control over what they consume and how they consume it.
- From outbound to inbound: consumers actively seek out products and services that meet their needs and preferences.

Network Effects

Network effects occur when the value of a platform increases or decreases for users as more participants join the platform. (Two-sided, different from price effects and brand effects)

- **Positive same side**: When the value of the platform increases for users on the same side as more users join (e.g. social networks like Facebook, where more friends joining increases the value for existing users).
- **Negative same side**: When the value of the platform decreases for users on the same side as more users join (e.g. congestion effects in ride-sharing platforms, where too many drivers can reduce earnings per driver).
- **Positive cross side**: When the value of the platform increases for users on one side as more users join the other side (e.g. more sellers on eBay attract more buyers, and vice versa).
- **Negative cross side**: When the value of the platform decreases for users on one side as more users join the other side (e.g. too many advertisers on a platform like YouTube can reduce the user experience for viewers due to excessive ads).

Monetization

- Be careful with charging: Overcharging can drive users away, while undercharging can lead to unsustainable operations. Striking the right balance is crucial.
- Different types of platform monetization:
  - **Transactional Fees**: Both buyers and sellers are not discouraged. Platforms should still provide other value added benefits to prevent users/producers taking the activity outside the platform
  - **Access/Enhanced Access**: Charge producers for access to consumers or charge users for lowering barriers.
  - **Enhanced Curation**: Improve quality of interactions of those who pay a premium. (e.g subscription for better vetting and screening of service providers)
- **Important principles:**
  - Avoid charging for value that users previously received for free.
  - Avoid reducing access to value that users have become accustomed to receiving.
  - Strive to create additional value that justifies the charge imposed.
  - Monetization strategies should be considered when making initial platform design choices.

Metrics

- **Metrics for Traditional Pipeline Model:**
  - Produce goods and services efficiently and sufficient for demand. (Operations)
  - Reach customers through proper channels at appropriate prices. (Marketing)
  - Generate adequate revenues to produce profits/value for investors (Finance)
  - E.g. Cash flow, inventory turns, operating income, gross margin, overhead, ROI
- *Platforms generate value through network effects*
- **Rate of interaction success** and contributing factors.
- Quantifying success in generating desirable interactions amongst participants.
- **Startup Phase:**
  - Growth of active producers and consumers who are participating in large volumes of successful interactions
  - **Liquidity**: The ease with which participants can find and engage with counterparties for successful interactions on the platform. (Active usage, active users $\div$ total users, new active users $\div$ total active users)

- **Matching Quality**: The degree to which the platform successfully connects participants with the right counterparties to fulfill their needs or preferences. (Percentage of search leading to interaction, correlation between interaction rate and long-term rate of activity)
- **Trust**: The level of confidence participants have in the platform and its participants to ensure fair and reliable interactions. (Review system)
- *Outcome based metrics, Metrics measuring commitment to ecosystem, Content creation metrics, Market access regardless of complete interaction*

- **Growth Phase:**
  - **Balance**: The platform's ability to maintain an appropriate ratio of producers to consumers to ensure a healthy ecosystem. (E.g. producer-to-consumer ratio, interaction success rate across different user groups)
  - **Side Switching Rate**: The frequency with which users switch roles between producers and consumers on the platform. (E.g. percentage of users who act as both producers and consumers within a given time period)
  - **Producer Side**
    - Frequency of producer participation, listings created, outcomes achieved
    - Interaction failure (fell through interactions)
    - Producer fraud (description and delivery)
    - Retention and Lifetime Value: customer retention rate, average customer lifetime value, churn rate [lower the better]
  - **Consumer Side Side:**
    - Frequency of consumption, searches, and rate of conversion to sale.
    - Retention and Lifetime Value metrics as in a traditional business.
- **Maturity Phase:**
  - Driving innovation: platform must adapt to the needs of its users and changes in competitive and regulatory environment.
  - E.g Incorporate third-party feature that provides a large share of value enjoyed by users as own feature.

Future of Platform Revolution

- **Joining the revolution:** Information intensive industries; Industries with non-scalable gatekeepers; Highly fragmented industries (market aggregation); Industries characterised by extreme information assymetries.
- **Resist disruption:** Industries with high regulatory control, Industries with high failure costs (patient-doctor); Resource intensive industries

Accounting Essentials

Balance Sheet

Assets	End of fiscal 2017	2016	Liabilities and shareholders' equity	End of fiscal 2017	2016
Current assets					
Cash and marketable securities	3,595	2,538	Current liabilities		
Receivables	1,952	2,029	Debt due for repayment	2,761	1,252
Inventories	12,748	12,549	Accounts payable	11,628	11,212
Other current assets	638	608	Other current liabilities	1,805	1,669
Total current assets	18,933	17,724	Total current liabilities	16,194	14,133
Fixed Assets			Long-term debt	24,267	22,349
Tangible fixed assets			Other long-term liabilities	2,614	2,151
Property, plant, and equipment	41,413	40,426			
Less accumulated depreciation	19,339	18,512	Total liabilities	43,075	38,633
Net tangible fixed assets	22,075	21,914			
			Shareholders' equity:		
Intangible asset (goodwill)	2,275	2,093	Common stock and other paid-in capital	9,715	9,010
Other assets	1,246	1,235	Retained earnings	39,935	35,517
			Treasury stock	(48,196)	(40,194)
			Total shareholders' equity	1,454	4,333
Total Assets	44,529	42,966			
			Total liabilities and shareholders' equity	44,529	42,966

Income Statement

	$ Million	% of Sales
Net sales	100,904	100.0%
Other income	325	0.3%
Cost of goods sold COGS	66,548	66.0%
Selling, general & administrative exper SGA	17,864	17.7%
Depreciation expense	2,062	2.0%
Earnings before interest and income ta	14,755	14.6%
Interest expense	1,057	1.0%
Taxable income	13,698	13.6%
Taxes	5,068	5.0%
Net income	8,630	8.6%
Allocation of net income		
Dividends	4,212	4.2%
Addition to retained earnings	4,418	4.4%

Statement of Cash Flow

Cash provided by operations: Add depreciation (it is not cash)	
Net income	8,630
Depreciation Don't include cash and short term debt	2,062
Changes in working capital items	
Decrease (increase) in accounts receivable	139
Decrease (increase) in inventories	(84)
Decrease (increase) in other current assets	(10)
Increase (decrease) in accounts payable	352
Increase (decrease) in other current liabilities	669
Total decrease (increase) in working capital	1,066
Cash provided by operations	11,758

Cash flows from investments:	
Capital expenditure	(1,897)
Sales (acquisitions) of long-term assets	47
Other investing activities	(105)
Cash provided by (used for) investments	(1,955)
Cash provided for (used by) financing activities:	
Increase (decrease) in short-term debt	850
Increase (decrease) in long-term debt	2,448
Dividends	(4,212)
Repurchases of stock	(7,745)
Other	(211)
Cash provided by (used for) financing activities	(8,870)
Net increase (decrease) in cash and cash equivalents	933

- Asset Decrease $\rightarrow$ Cash Inflow
- Asset Increase $\rightarrow$ Cash Outflow
- Liability Increase $\rightarrow$ Cash Inflow
- Liability Decrease $\rightarrow$ Cash Outflow

Candy Canes Inc. spends $100,000 to buy sugar and peppermint in April. It produces its candy and sells it to distributors in May for $150,000, but it does not receive payment until June. Assuming that sales in April and June are zero, fill in the following table.

Candy Canes Inc.	April	May	June
Sales	$ 0	$ 150,000	$ 0
Cash flow*	(100,000)	0	150,000
Net income**	0	50,000	0

Notes

- **Interest expense (from loans) is tax exempt**. If the company is fully financed by equity, the tax shield benefit of debt financing is not utilized, potentially leading to higher overall tax liabilities.
- If the firm has Net Operating Loss (negative taxable income), it can carry forward the loss to offset taxable income in future years, reducing future tax liabilities. This is known as a **tax loss carryforward**. For example:
  - Year 1: Net operating loss of $100,000. Year 2: Taxable income of $150,000. Loss carryforward reduces taxable income to $50,000, lowering Year 2 taxes.
- **Marginal tax rate:** highest tax rate a taxpayer encounters
- **Notes payable:** is a short-term debt and current liability if the repayment terms is due within 1 year.

Formulas and Equations

- assets $=$ liability $+$ equity
- equity $=$ common stock $+$ treasury stock $+$ retained earnings
- net working capital $=$ current assets $-$ current liabilities
- EBIT $=$ total revenue $+$ other income $-$ COGS $-$ depreciation
- taxable income $=$ EBIT $-$ interest expense
- net income $=$ taxable income $-$ taxes
- taxes $=$ taxable income [EBT] $\times$ tax rate
- retained earnings $=$ net income $-$ dividends
- net increase (decrease) in cash and equivalents $=$ cash operations $\pm$ cash investments $\pm$ cash financing activities
- free cash flow $=$ net income $+$ interest $+$ depreciation $-$ additions to networking capital $+$ cashflow from investment
- free cash flow $=$ interest $+$ cashflow from operations $+$ cashflow from investments (aka capital expenditure)
- earnings per share $= \frac{\text{net income}}{\text{no. of outstanding shares}}$
- average tax bracket [tax rate] $= \frac{\text{taxes}}{\text{taxable income}}$
- cash flow $=$ net income $+$ depreciation

Financial Statement Analysis

Notes

- **Market value** fluctuates due to changes in investor sentiment, market conditions, and company performance.
- We are unable to look up the **market value of privately owned companies** or divisions/plants of larger companies because their shares are not publicly traded.
- **Cost of Capital:** The minimum return that investors expect for providing capital to the company, reflecting the opportunity cost of investing elsewhere.
- **Economic value added (EVA)/Residual Income:** Net dollar return generated by the company after deducting the cost of capital from its operating profit. Positive EVA indicates the company is creating value for shareholders, while negative EVA indicates value destruction.
- Advantages of **EVA:** Recognises companies need to cover opportunity costs before they add value. Makes cost of capital visible, managers need to earn at least cost of capital. Manager can improve EVA by reducing assets that are not contributing to profits.
- Disadvantages of **EVA:** Shows current performances and not affected by other things that move the stock market. Book values used.
- **Compare cost of capital (expectation) with return on capital (actual)**. If ROC $>$ COC, the company is generating profits above the expected return, creating value for shareholders (EVA will be positive). Conversely, if ROC $<$ COC, the company is destroying value as it is not meeting the expected return (EVA will be negative).
- **After-tax operating income (ATOI)**: Net income had the company been financed entirely with equity, excluding the effects of interest expenses and tax shields from debt financing.
- **ATOI** is calculated before interest expense, while net income is calculated after deducting interest expense. ATOI reflects the company's operating performance regardless of its financing structure, whereas net income accounts for the impact of financing decisions.
- When considering **mergers**, consider the sales, profits, assets, asset turnover, profit margin and ROA.
- Industries with high asset turnover often have lower profit margins but rely on high sales volume to generate profits. Vice versa.
- **ROA** is independent of financial leverage. If ROA exceeds the cost of debt, leveraging can increase ROE.
- Generally, we want **current ratio** to be $>$ 1. But some companies (e.g universities) have high unearned revenue, increasing current liabilities.

Formulas and Equations

- market capitalization $=$ share price $\times$ no. of outstanding shares
- market value added $=$ market capitalization $-$ book equity value
- market to book ratio $=$ market equity value $\div$ book equity value
- market to book ratio $=$ market price per share $\times$ book equity value per share
- increase in equity investment value $=$ market value added $\div$ book equity value
- EVA $=$ ATOI $-$ (cost of capital $\times$ total capitalization)
- ATOI $= (1 -$ tax rate$) \times$ interest expense $+$ net income
- ATOI $=$ EBIT $\times (1 -$ tax rate$)$
- total capitalization $=$ long-term debt $+$ equity [only]
- return on capital (ROC) $= \frac{\text{after-tax operating income}}{\text{total capitalization}}$
- return on assets (ROA) $= \frac{\text{after-tax operating income}}{\text{total assets}}$
- return on equity (ROE) $= \frac{\text{net income}}{\text{equity}}$
- asset turnover $= \frac{\text{sales}}{\text{total assets (start of year)}}$ or $\frac{\text{sales}}{\text{average total assets}}$
- total sales $=$ receivables or cash $\times \frac{365}{\text{no. of days representing sales}}$
- inventory turnover $= \frac{\text{COGS}}{\text{inventory (start of year)}}$
- average days in inventory $= \frac{\text{inventory (start of year)}}{\text{COGS} \div 365}$
- receivables turnover $= \frac{\text{sales}}{\text{receivables (start of year)}}$
- average collection period $= \frac{\text{receivables (start of year)}}{\text{sales} \div 365}$
- average payment delay $= \frac{\text{accounts payable (start of year)}}{\text{COGS} \div 365}$
- profit margin $= \frac{\text{net income}}{\text{sales}}$
- operating profit margin $= \frac{\text{after-tax operating income}}{\text{sales}}$
- ROA $=$ asset turnover $\times$ operating profit margin $= \frac{\text{sales}}{\text{total assets}} \times \frac{\text{ATOI}}{\text{sales}}$
- long-term debt ratio $= \frac{\text{long-term debt}}{\text{long-term debt + equity}} = \frac{\text{long-term debt}}{\text{total capitalization}}$
- long-term debt-equity ratio $= \frac{\text{long-term debt}}{\text{equity}}$
- total debt ratio $= \frac{\text{total liabilities}}{\text{total assets}}$
- times interest earned $= \frac{\text{EBIT}}{\text{interest payments}}$
- cash coverage ratio $= \frac{\text{EBIT+depreciation}}{\text{interest payments}}$
- ROE (DuPont) $=$

$$\underbrace{\frac{\text{assets}}{\text{equity}}}_{\text{leverage ratio}(\geq 1)} \cdot \overbrace{\underbrace{\frac{\text{sales}}{\text{assets}}}_{\text{asset turnover}} \cdot \underbrace{\frac{\text{ATOI}}{\text{sales}}}_{\text{operating profit margin}}}^{\text{ROA}} \cdot \underbrace{\frac{\text{net income}}{\text{ATOI}}}_{\text{debt burden}(\leq 1)}$$

- cost of debt $=$ interest rate $\times (1 -$ tax rate$)$
- net working capital to total assets ratio $= \frac{\text{net working capital}}{\text{total assets}}$
- current ratio $= \frac{\text{current assets}}{\text{current liabilities}}$
- quick ratio $= \frac{\text{cash\&marketable securities+receivables}}{\text{current liabilities}}$ [no inv]
- cash ratio $= \frac{\text{cash\&marketable securities}}{\text{current liabilites}}$